

INTA 2040

Science, Technology, and International Affairs

Week-by-Week Class Schedule

Term: Fall 2026

Credit Hours: 3

Time: Mondays & Wednesdays, 12:30–1:45 pm

Location: Habersham G17

Instructor Information

Instructor: Dr. Juljan Krause

Office: Habersham 315

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Office Hours: Wednesdays, 2:00–4:00 pm

How to use this document

This document has a twin

This document is the Course Schedule that presents an overview of what we are doing each week, what to read, and when work is due. The **INTA 2040 Syllabus on Canvas** contains the course description and learning outcomes, grading and assessment details, and all course policies.

Please read the two together. The syllabus is the stable reference policy document; this schedule is the working document for the semester and may be updated as the course develops. They work in tandem in that the syllabus governs on policy and grading, while this schedule governs on dates, topics, and readings.

A note on changes to this document

This course deals with technologies and policies that evolve quickly at times. If something important happens during the semester, I may substitute elements of a topic or reading so that we can discuss it while it matters. Changes, if any, will be to content, not to the basic structure of the course.

It is important that you are able to plan around the dates, deadlines, and assessment weightings in this schedule, so these will not change unless Institute-wide emergency measures beyond the instructor's control demand it. Any changes will be announced on Canvas and by email with as much notice as possible.

At a Glance

Non-meeting dates: no classes on

Monday 7 September Labor Day

Monday 5 October Fall Recess

Wednesday 25 November Thanksgiving break

Wednesday 9 December No regular class; our final class is Monday 7 December

Assessment calendar

Reading quizzes (15%) 11 quizzes, lowest 2 dropped. Unless Canvas says otherwise, a quiz is due at 12:00 pm before the first class meeting of the week in which it appears below. There are no quizzes in Weeks 1, 3, 8, or 15.

Event reflection Friday 11 September. This is an engagement activity and contributes to Participation & Engagement.

Participation & engagement (10%) Continuous, including in-class exercises and the Week 3 event reflection.

Midterm exam (20%) Wednesday 14 October; covers Weeks 1–7.

Analytical essay (25%) Friday 6 November, 5:00 pm.

Final exam (30%) Wednesday 16 December, 11:20 am–2:10 pm, according to the current Georgia Tech Fall 2026 Final Exam Matrix.

Please see the INTA 2040 Syllabus on Canvas for full descriptions of the course rationale, each graded component, and the grading scale.

Getting the most out of the readings

There is no required textbook. Most weeks pair one analytical reading with one current empirical or primary-policy reading. Required reading will normally remain within the syllabus expectation of approximately 40–60 pages per week; when a book excerpt, report, or treaty is longer, the assigned selection will be identified on Canvas. Recommended items are always optional.

The links below point to public, official, open-access, or publisher pages where possible. Canvas remains the authoritative source for the exact excerpt and for any licensed or accessibility-adjusted copy.

Week 1 | Why is technology a subject for international affairs?

Monday 24 August | Wednesday 26 August

Monday: Course launch. Our overarching question: how technology shapes power, and how power shapes technology. Course mechanics and logistics.

Wednesday: Core concepts: general-purpose technologies, dual-use, innovation versus diffusion. Why some technologies alter strategic competition and others do not. We distinguish technical novelty from strategic consequence and ask what evidence would let us tell the difference.

Required readings

- Horowitz, M. C. (2020). *Do Emerging Military Technologies Matter for International Politics?*. *Annual Review of Political Science* 23, pp. 385–400.
- Ding, J. & Dafoe, A. (2023). *Engines of Power: Electricity, AI, and General-Purpose Military Transformations*. *European Journal of International Security* 8(3), pp. 377–394.

No quiz in week 1. We'll run through the syllabus and weekly schedule together in class.

Week 2 | Chokepoints, actors, and technology statecraft

Monday 31 August | Wednesday 2 September

Monday: Network effects and weaponized interdependence: why control over a network node can matter more than nominal ownership of a technology. Who actually makes technology policy? We'll discuss states, firms, international institutions, scientific communities, universities, and standards bodies.

Wednesday: Introducing the analytical framework of the course. Over the coming weeks, we'll ask five questions of every technology: (1) What is it and how does it work? (2) Why is it strategically significant? (3) Who are the key actors? (4) What governance exists? (5) What are the core tensions and trade-offs?

We'll discuss three postures states often adopt toward science and technology (promoting capability, protecting capability, and projecting influence) and the instruments they typically choose: R&D systems, talent policy, research security, industrial policy, export controls, standards, and science diplomacy.

Required readings

- Farrell, H. & Newman, A. L. (2019). *Weaponized Interdependence: How Global Economic Networks Shape State Coercion*. *International Security* 44(1), pp. 42–79.

- Bradford, A. (2012). *The Brussels Effect*. Northwestern University Law Review, 107(1), [READ pp. 1-19 only]

Recommended / optional

- National Science Foundation. *Research Security Training*. Current U.S. research-security requirements and training materials.

Due: Quiz 1, Monday 12:00 pm before class.

Week 3 | Intelligence, technology, and statecraft

Wednesday 9 September | Monday 7 September is Labor Day, no class

Special session: the class attends the CIA AI Director's address as a body. There is no regular classroom meeting this week.

Before: Post two questions to the Canvas discussion by Tuesday 8 September, each mapped to one of our five framework questions. Arrive with something specific to listen for, and feel encouraged to ask a question (no penalties if you don't, events of this kind can feel somewhat intimidating.)

After: A short reflection of 400–500 words, due Friday 11 September. *Identify one claim the Director made about technology and national security. Which of our five questions does it answer, if any? What would you need to know to assess whether it is right?* This is part of Participation & Engagement rubric.

Required readings

- Office of the Director of National Intelligence (2026). *Annual Threat Assessment of the U.S. Intelligence Community*. Read the sections most directly concerned with technological competition, cyber, space, AI, and other emerging technologies.
- Central Intelligence Agency. *Technology Collaboration*. A short official overview of how the Agency works with commercial technology and venture-backed innovation, including In-Q-Tel.

Due: Event reflection, Friday 11 September. No reading quiz this week.

Week 4 | The original strategic technology

Monday 14 September | Wednesday 16 September

Monday: From Manhattan to Hiroshima. How a scientific program became a state capability, and the institutional requirements it entailed: concentrated resources, secrecy, scientific labor, industrial capacity, and political authority.

Wednesday: The logic of nuclear deterrence: second strike, credibility, escalation, and the stability–instability paradox.

Required readings

- Schelling, T. C. (1966). *Arms and Influence*. Yale University Press, Ch. 1, “The Diplomacy of Violence.”
- Cirincione, J. (2007). *Bomb Scare: The History and Future of Nuclear Weapons*. Columbia University Press, Introduction and Ch. 1.

Due: Quiz 2, Monday 12:00 pm.

Week 5 | Governing the bomb

Monday 21 September | Wednesday 23 September

Monday: The nonproliferation architecture: NPT, IAEA safeguards, export-control regimes, and verification. The inseparable relationship between engineering and governance: the CTBTO’s seismic, hydroacoustic, infrasound, and radionuclide monitoring, and the IAEA’s use of environmental sampling.

Wednesday: Norms and taboos as governance. What does nuclear governance actually teach us about AI, cyber, biotechnology, quantum, or space and where does the analogy break down?

Required readings

- Tannenwald, N. (1999). *The Nuclear Taboo: The United States and the Normative Basis of Nuclear Non-Use*. *International Organization* 53(3); selected excerpt rather than the full article.
- United Nations (1968). *Treaty on the Non-Proliferation of Nuclear Weapons*. Full text — shorter than it looks.
- Kosal, M. E. (2021). *CRISPR and New Genetic Engineering Techniques: Emerging Challenges to Strategic Stability and Nonproliferation*. *The Nonproliferation Review* 27(4–6), pp. 389–408.

Recommended / optional

- Comprehensive Nuclear-Test-Ban Treaty Organization. *International Monitoring System*. How the global verification network works.

- International Atomic Energy Agency. [IAEA Safeguards Overview](#). Reference material on verification and safeguards.

Due: Quiz 3, Monday 12:00 pm.

Block III: The International Technology Landscape

Weeks 6–13

Week 6 | AI I: What is AI really?

Monday 28 September | Wednesday 30 September

Monday: The key features of modern AI systems: training versus inference, transformers and attention, scaling, data, compute, benchmarks, and the slippery concept of “capability.” We ask why benchmark performance can improve faster than our confidence that the benchmark measures what actually really matters.

Wednesday: Seerist lab & AI practice.

A hands-on trial of a commercial AI-augmented threat and risk intelligence platform. Seerist ingests open-source material at scale, produces risk assessments, and combines algorithmic monitoring with the input of human analysts.

We’ll run Seerist in small groups in class together for you to develop a sense of how these platforms work, and where the critical human oversight is (still) required.

Required readings

- Mitchell, M. (2021). [Why AI is Harder Than We Think](#). arXiv:2104.12871.
- Stanford Institute for Human-Centered Artificial Intelligence (2026). [AI Index 2026: Technical Performance](#). Selected sections on benchmarks, reasoning, agents, model convergence, and the changing performance frontier.

Recommended / optional

- Bommasani, R. et al. (2021). [On the Opportunities and Risks of Foundation Models](#). Stanford CRFM / arXiv.
- Vaswani, A. et al. (2017). [Attention Is All You Need](#). The paper behind the architecture; not required for technical mastery.

Due: Quiz 4, Monday 12:00 pm.

Week 7 | AI II: Great-power competition and the material bases of AI power

Wednesday 7 October | Monday 5 October is Fall Recess, no class

Wednesday: “AI race” rhetoric vs. reality. Innovation versus diffusion; compute, data, talent, capital, and energy as strategic resources; the geography of data centers and the grid as part of AI infrastructure; and the relationship between frontier firms and states. We compare contemporary U.S. and Chinese strategy.

Required readings

- Ding, J. (2024). *Technology and the Rise of Great Powers: How Diffusion Shapes Economic Competition*. Princeton University Press, Introduction, pp. 1–14.
- Sastry, G. et al. (2024). *Computing Power and the Governance of Artificial Intelligence*. Selected introductory sections on compute as a governable resource.
- **Current-policy packet:** The White House (2025), *America’s AI Action Plan*, selected sections on innovation and infrastructure; paired with the Chinese Ministry of Foreign Affairs (2025), *Global AI Governance Action Plan*, selected sections. Read the short extracts posted on Canvas rather than either document in full.

Due: Quiz 5, Wednesday 12:00 pm.

Week 8 | AI III: Governance, regulation, and autonomous systems

Monday 12 October | Wednesday 14 October

Monday: Can AI be governed? We compare governance at different layers: models, compute, deployment, liability, standards and sectoral rules, using the chokepoint logic from Week 2, and the norms-and-verification problem from Week 5. The EU AI Act, the contemporary U.S. standards-and-innovation architecture, and international approaches offer very different answers to a shared problem. We then turn to autonomous weapons: meaningful human control, verification, proliferation, and whether nuclear-era arms-control concepts still work.

A note on the midterm. Wednesday’s exam covers Weeks 1–7. Today is therefore primarily revision and no new concepts are introduced, only a new case to illustrate what we’ve discussed so far.

Wednesday: MIDTERM EXAM. Short-answer questions and one essay. Covers Weeks 1–7. No new material.

Required readings

- Scharre, P. (2018). *Army of None: Autonomous Weapons and the Future of War*. Selected chapter on autonomy and control.
- **Current-policy packet:** European Commission, *AI Act regulatory framework* (current implementation overview); NIST, *Center for AI Standards and Innovation*; and the UN CCW, *Guiding Principles on Lethal Autonomous Weapons Systems*. Read the short selections posted on Canvas rather than the full legal instruments.

Due: No reading quiz due to the joy that is your **MIDTERM EXAM on Wednesday 14 October.**

Week 9 | Semiconductors, supply chains, and industrial policy

Monday 19 October | Wednesday 21 October

Monday: Why everyone is talking about the semiconductor stack: chip design, EDA, fabrication, equipment, materials, advanced packaging, and leading-edge nodes. Extreme-ultraviolet lithography is a great case study for us: tin droplets, laser-generated plasma, multilayer mirrors, and a hyper-specialized industrial ecosystem that have produced chokepoints that can become geopolitically consequential.

Wednesday: Export controls and industrial policy as statecraft. The evolution of allied controls, subsidies, friend-shoring, localization, and the possibility that denial accelerates domestic substitution. Who bears the cost of a chokepoint strategy, and what happens when allies control, or don't control, crucial nodes?

Required readings

- Miller, C. (2022). *Chip War: The Fight for the World's Most Critical Technology*. Scribner, selected chapters on the contemporary semiconductor production network.
- Center for Strategic and International Studies (2026). *China's Localization Drive in Semiconductors Gains Impetus from Allied Chip Export Controls*. Current assessment of controls, localization, and substitution.
- OECD (2025). *Reconfiguring Scientific Co-operation in a Changing Geopolitical Environment*. *OECD Science, Technology and Innovation Outlook 2025*; selected sections on promotion, protection, projection, research security, and science diplomacy. This is the Week 2 map, now applied to a live chokepoint.

Due: Quiz 6, Monday 12:00 pm.

Week 10 | Internet architecture and governance

Monday 26 October | Wednesday 28 October

Monday: Why the internet is one of the most important technologies in history (if not the most important one). How it works and who governs what: autonomous systems, routing, DNS, ICANN, IETF, and ITU. We use BGP and route hijacking as the technical case to illustrate why a protocol built on inter-network trust can create strategic vulnerability.

Wednesday: Multistakeholder versus multilateral governance, data localization, sovereignty, technical standards, and fragmentation. The UN Global Digital Compact lets us ask how states outside the familiar U.S.–China–EU triangle frame openness, interoperability, development, and governance. Is the “splinternet” happening, and what evidence would demonstrate it?

Required readings

- DeNardis, L. (2020). *The Internet in Everything: Freedom and Security in a World with No Off Switch*. Yale University Press, Ch. 1.
- United Nations (2024). *Global Digital Compact*. Selected principles and commitments on internet governance, interoperability, digital divides, and international cooperation.

Recommended / optional

- Hoffmann, S., Lazanski, D. & Taylor, E. (2020). *Standardising the splinternet: how China's technical standards could fragment the internet*. *Journal of Cyber Policy* 5(2).
- Cloudflare. *BGP and Route Security*. A current technical explainer.

Due: Quiz 7, Monday 12:00 pm.

Week 11 | Cybersecurity and conflict in cyberspace

Monday 2 November | Wednesday 4 November

Monday: What a state cyber operation actually involves: access, persistence, privilege, command and control, and effects. We discuss Stuxnet as a case of code crossing into physical process manipulation; critical-infrastructure cases then show why defense is partly an organizational and governance problem.

Wednesday: Deterrence, restraint, and international norms. What has changed since the first UN GGE agreements? We use the 2025 Open-Ended Working Group final report to examine the present normative architecture, confidence-building, capacity-building, and the problem of translating broad commitments into actual state behavior.

Required readings

- Buchanan, B. (2017). *The Cybersecurity Dilemma: Hacking, Trust and Fear Between Nations*. Oxford University Press, Introduction and selected sections of Ch. 2.
- United Nations (2025). *Final report of the Open-Ended Working Group on security of and in the use of information and communications technologies 2021–2025*. A/80/257; selected sections on norms, confidence-building, capacity-building, and future mechanisms.

Recommended / optional

- Nye, J. S. (2017). *Deterrence and Dissuasion in Cyberspace*. *International Security* 41(3), pp. 44–71.
- Falliere, N., O Murchu, L. & Chien, E. (2011). *W32.Stuxnet Dossier*. Symantec, Version 1.4; technical reference used in class.

Due: Quiz 8, Monday 12:00 pm. **ANALYTICAL ESSAY DUE Friday 6 November, 5:00 pm.**

Week 12 | Quantum technologies and the cryptographic transition

Monday 9 November | Wednesday 11 November

Monday: Quantum computing, sensing, and communications: three technology families that are often wrongly collapsed into one umbrella term. We separate what exists now from what requires further engineering, and ask how to reason about strategic significance under extreme uncertainty and hype.

Wednesday: The cryptographic transition. Why Shor's algorithm threatens RSA and elliptic-curve cryptography specifically rather than making computers generically "faster"; what "harvest now, decrypt later" means; how NIST's post-quantum standards such as ML-KEM change the problem; and why migration across large and complex computer systems is a massive governance and infrastructure challenge, not just an algorithm-selection exercise.

Required readings

- U.S. Government Accountability Office (2026). *Quantum Computing: Leadership Needed to Coordinate Federal Efforts*. GAO-26-107759; selected sections on technology status, risks, and federal strategy.
- **Current-policy packet:** NIST, *Migration to Post-Quantum Cryptography* (selected guidance), paired with the White House, *Securing the Nation Against Advanced Cryptographic Attacks* (2026), selected sections.

Recommended / optional

- Krause, J. (2026). *Salt Typhoon and the Limits of "Quantum Security": A Technique-Level Audit*. *IEEE Security & Privacy* 24(4), pp. 59–69.

Due: Quiz 9, Monday 12:00 pm.

Week 13 | Space and dual-use systems

Monday 16 November | Wednesday 18 November

Monday: Space as dual-use infrastructure: position, navigation and timing; ISR; satellite communications; launch; ground stations; and commercial providers. GPS and clocks, signals, constellations, and dependence on PNT. Then we ask why civilian and military infrastructures can become strategically entangled.

Wednesday: Counterspace capabilities and governance. Reversible versus irreversible effects, attribution, debris, commercial entanglement, and a thin legal regime.

Required readings

- Bowen, B. E. (2020). *War in Space: Strategy, Spacepower, Geopolitics*. Edinburgh University Press, selected chapters.
- Secure World Foundation (2026). *Global Counterspace Capabilities*. Executive Summary and selected country/capability sections.

- United Nations (1967). *Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space*. Outer Space Treaty, full text.

Recommended / optional

- GPS.gov. *The Global Positioning System*. Technical primer on the space, control, and user segments.

Due: Quiz 10, Monday 12:00 pm.

Block IV: Synthesis

Weeks 14–15

Week 14 | Patterns across cases: governing openness, security, and interdependence

Monday 23 November | Wednesday 25 November is Thanksgiving break, no class

Monday: What has actually recurred over the past weeks: concentrated chokepoints versus diffuse capability; observability and verification; openness versus research security; innovation versus control; alliance dependence; firms as strategic actors; and the speed of technology against the speed of institutions.

Required readings

- UK House of Commons Science, Innovation and Technology Committee (2026). *Science diplomacy: sovereignty, strategy, and the global race*. Selected summary and analytical sections on science diplomacy under strategic competition.
- OECD (2025). *Reconfiguring Scientific Co-operation in a Changing Geopolitical Environment*. Revisit the promotion–protection–projection framework and the science-diplomacy sections from Week 2.

Recommended / optional

- European Commission (2026). *A European Framework for Science Diplomacy*. Current EU framework and background materials.

Due: Quiz 11, Monday 12:00 pm.

Week 15 | Firms, futures, and the international system

Monday 30 November | Wednesday 2 December | Monday 7 December

Monday 30 November: More synthesis: corporate power and the technopolar question. What happens to sovereignty when frontier capability, infrastructure, standards, data, or operational expertise sits with firms?

Wednesday 2 December: What does strategic competition over emerging technologies mean for the international system? We revisit the Week 5 nuclear-governance framework and ask if

it is fit for purpose today.

Monday 7 December: We apply the five-question framework to a technology we never covered. No special factual knowledge is required; rather the idea is to identify what we would need to know, trace technical properties into strategic consequences, map actors and governance, and state the core trade-offs. This is for us to practice applying an INTA 2040 framework to something new and different outside the classroom.

Required readings

- Bremmer, I. (2021). *The Technopolar Moment: How Digital Powers Will Reshape the Global Order*. Foreign Affairs.
- Danzig, R. (2018). *Technology Roulette: Managing Loss of Control as Many Militaries Pursue Technological Superiority*. Center for a New American Security; Executive Summary and selected sections.

Recommended to revisit

- Farrell, H. & Newman, A. L. (2019). *Weaponized Interdependence: How Global Economic Networks Shape State Coercion*. Return to your Week 2 notes and the paper's core mechanism with the full course now behind it.

Due: No quiz, happy days! Alas, the **FINAL EXAM, most likely on Wednesday 16 December, 11:20 am–2:10 pm** (current Georgia Tech Fall 2026 Final Exam Matrix).

This schedule should be read alongside the INTA 2040 Syllabus on Canvas, which contains a detailed course description, learning outcomes, full assessment descriptions, grading scale, and all course policies.